

Toronto Building Permits Data Analysis

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Table of Contents

Chapter 1. Problem Definition	7
1.1 Background.....	7
1.2 Research Questions	7
1.3 Justification of Problem Definition and Research Questions	8
Chapter 2: Hypotheses Formulation.....	9
2.1 Hypothesis 1 (H_1).....	9
2.2 Hypothesis 2 (H_2).....	9
2.3 Hypothesis 3 (H_3).....	9
Chapter 3. Data Collection and SQL Queries.....	10
3.1 Data Source and Database Structure.....	10
3.2 SQL Techniques and Extraction Approach	11
3.3 Data Extraction for Research Question 1	11
3.4 Data Extraction for Research Question 2.....	11
3.5 Data Extraction for Research Question 3.....	12
Chapter 4. Data Understanding	13
4.1 Overview of Analytical Datasets	13
4.2 Descriptive Analysis for Research Question 1	13
4.2.1 Exploratory Analysis and Statistical Tests of RQ 1.....	14
4.3 Descriptive Analysis for Research Question 2	15
4.3.1 Exploratory Analysis and Statistical Tests of RQ 2.....	16
4.4 Descriptive Analysis for Research Question 3	16

4.4.1 Exploratory Analysis and Statistical Tests of RQ 3.....	17
Chapter 5. Data Visualization.....	19
5.1 Visualization Approach	19
5.2 Visualizations for Research Question 1	19
5.3 Visualizations for Research Question 2	21
5.4 Visualizations for Research Question 3	22
Chapter 6: Model Building	25
6.1 Data Preparation for Modeling	25
6.2 Predictive Model to Address Research Question 1	25
6.3 Predictive Model to Address Research Question 2.....	27
6.4 Predictive Model to Address Research Question 3.....	29
6.5 Summary of Model Development.....	31
Chapter 7: Model Evaluation	33
7.1 Evaluation of Model for Research Question 1: Estimated Construction Cost.....	33
7.2 Evaluation of Model for Research Question 2: Estimated Permit Approvals	34
7.3 Evaluation of Model for Research Question 3: Estimated Dwelling Units Created.....	35
7.4 Assessment of Model Performance Against Benchmarks	36
7.5 Strengths, Limitations, and Implications of the Models	37
Chapter 8. Conclusion	38
References.....	40

Table of Tables

Table 1.....	14
Table 2.....	15
Table 3.....	17
Table 4.....	27
Table 5.....	29
Table 6.....	31
Table 7.....	31
Table 8.....	33
Table 9.....	33
Table 10.....	35
Table 11.....	35

Table of Figures

Figure 120

Figure 221

Figure 322

Figure 423

Figure 524

Table of Appendix

Appendix A.....	41
Appendix B.....	43
Appendix C.....	45
Appendix D.....	47
Appendix E.....	48
Appendix F.....	50
Appendix G.....	53
Appendix H.....	54
Appendix I.....	55

Chapter 1. Problem Definition

1.1 Background

Over the last decade, Toronto has undergone rapid expansion, emerging as one of the fastest-growing cities in Canada in both residential and commercial development. According to Urbanize Toronto (2024), the city's population increased by more than 220,000 people in the year ending July 2023, making Toronto the fastest-growing metropolitan area in North America.

This growth has intensified activity in the construction sector, a vital part of Toronto's economy that requires strong oversight to ensure sustainable and equitable urban development. The city's Building Permit system serves as a key regulatory mechanism, tracking and approving construction activities across various neighborhoods.

The Toronto Building Permits dataset provides detailed insights into thousands of projects, including information on permit type, structure type, work type, and geographical grid areas. Analyzing this data allows researchers to identify trends in construction costs, project distribution, and administrative patterns.

This study aims to explore the factors that drive construction costs in Toronto and how these costs differ across neighborhoods. Beyond economics, the research seeks to uncover whether the city's development is distributed equitably, contributing to informed urban planning and fair policy design.

1.2 Research Questions

How do construction costs vary across different structure types in Toronto?

Are there significant differences in permit approval rates among Toronto's geographical grid areas?

How does the type of construction work relate to the number of dwelling units created?

1.3 Justification of Problem Definition and Research Questions

The problem is justified by Toronto's unprecedented population growth and urban expansion, which have significantly increased construction activity and housing demand. Understanding how costs and permit approvals differ by structure type and ward is crucial for identifying inefficiencies and ensuring equitable development across the city. The research questions were developed to provide insight into key drivers of construction trends and policy implications for urban sustainability.

Chapter 2: Hypotheses Formulation

2.1 Hypothesis 1 (H_1)

Null Hypothesis (H_{01}): Estimated construction cost does not differ significantly across structure types.

Alternative Hypothesis (H_{11}): Estimated construction cost differs significantly across structure types due to differences in design complexity, materials, and project scope.

2.2 Hypothesis 2 (H_2)

Null Hypothesis (H_{02}): The probability of permit approval does not differ across geographical grid areas.

Alternative Hypothesis (H_{12}): The probability of permit approval differs across geographical grid areas, influenced by local administrative efficiency and project volume.

2.3 Hypothesis 3 (H_3)

Null Hypothesis (H_{03}): Work type has no effect on the number of dwelling units created.

Alternative Hypothesis (H_{13}): The type of construction work has a direct impact on the number of dwelling units created—new constructions tend to increase housing supply, while renovations and alterations result in minimal change.

Chapter 3. Data Collection and SQL Queries

The present chapter explains how the datasets used to answer the research questions were extracted from the database. In line with the project rubric, it documents (1) the data source and database structure, (2) the SQL techniques applied, and (3) the specific SQL queries that were executed to obtain the analytical tables. All queries were run on the TorontoBuildingPermits MySQL database provided by the instructor.

3.1 Data Source and Database Structure

The dataset used in this project originates from the Toronto Building Permits MySQL database, which contains administrative records of construction permits issued across the City of Toronto. The database includes variables related to estimated construction cost, structure characteristics, permit status, work performed, geographical location, and changes in dwelling units.

The central table in the database is permit, which contains one record per permit and includes foreign keys referencing several supporting dimension tables. The most relevant tables used in this analysis include:

- permit – main dataset with estimated cost, dwelling units created/lost, status, structure type, work type, and ward.
- structure_type – categorical information about building structure types.
- work_type – classification of the type of construction work performed.
- permit_status – definitions of the current permit status (e.g., Permit Issued, Inspection, Revision Issued).
- ward – geographical information for Toronto's ward grids and names.

Together, these tables supply all the variables required to address the three research questions on construction costs, permit approvals, and dwelling-unit creation.

3.2 SQL Techniques and Extraction Approach

To transform the raw permit records into analytical datasets, several SQL operations were applied consistently across the three research questions. The main techniques included:

- SELECT to retrieve specific variables.
- JOIN to merge records across related tables.
- GROUP BY to aggregate measures at the structure type, work type, or ward level.
- WHERE to filter incomplete or irrelevant records.
- Aggregation functions such as COUNT, AVG, SUM, MIN, MAX, and COALESCE to compute descriptive statistics.

These techniques enabled the transformation of the raw permit-level records into structured analytical datasets aligned with the project's research questions.

3.3 Data Extraction for Research Question 1

RQ1: How do construction costs vary across different structure types in Toronto?

To address this question, two SQL queries were executed. The first query identifies all structure types available in the dataset. (Appendix A).

The second query computes summary statistics of estimated construction cost for each structure type, excluding records where the cost field is missing (`est_const_cost IS NULL`) (Appendix A).

3.4 Data Extraction for Research Question 2

RQ2: Do permit approval rates differ across Toronto's ward grids?

To answer this question, two SQL queries were executed. The first query retrieves the list of all permit status categories available in the dataset. (Appendix B).

The second query aggregates permits at the ward-grid level and computes each ward's approval rate, classifying "*Permit Issued*" and "*Revision Issued*" as approved outcomes. (Appendix B).

3.5 Data Extraction for Research Question 3

RQ3: How does the type of construction work relate to the number of dwelling units created?

To investigate this relationship, two SQL queries were used. The first query retrieves the list of all work types included in the dataset. (Appendix C).

The second query calculates several key measures for each work type: the total number of permits issued, the total number of dwelling units created, the average number of units created per permit, and the net change in dwelling units, computed as units created minus units lost. These metrics help identify which work types contribute most significantly to overall housing production. (Appendix C).

Chapter 4. Data Understanding

Chapter 4 reports the descriptive analysis of the datasets extracted in Chapter 3. The goal is to summarize key metrics, highlight main patterns, and provide an overall understanding of the data before moving to visualization and modeling. All descriptive statistics were produced in Excel using the result sets exported from MySQL.

4.1 Overview of Analytical Datasets

Three analytical tables were created, one for each research question; For RQ1, the data were aggregated by structure type to obtain construction cost measures; For RQ2, the data were aggregated by ward to obtain permit approval measures; For RQ3, the data were aggregated by work type to obtain dwelling-unit measures. In each case, records with missing or invalid key values were excluded so that the summaries are based on consistent, reliable information.

4.2 Descriptive Analysis for Research Question 1

RQ1 examines how estimated construction costs vary across structure types. For each structure type, Excel was used to calculate the number of permits, the average estimated construction cost, and the total construction value. Table 1 summarizes these metrics. The results show substantial variation across structure types: some categories combine relatively few permits with very high average costs, while others show many smaller projects with lower average costs. This pattern indicates important differences in project size and investment across structure types and motivates further analysis of Hypothesis 1.

Table 1

Descriptive statistics for estimated construction cost by structure type

StructureType	NumberOfPermits	AvgConstructionCost	TotalConstructionCost
Mixed Use/Res w Non Res	252	36817873.93	9278104229
Apartment Building	1524	3810268.251	5806848815
Transit Station,Subway, Bus Terminal	177	14386998.81	2546498789
Office	2873	561392.9808	1612882034
Multiple Use/Non Residential	830	1553876.304	1289717333
Hospital	140	7266248.979	1017274857
Multiple Unit Building	1328	671904.1911	892288765.8
Long Term Care Facility	47	7443166.685	349828834.2
Other	407	692503.1773	281848793.2
SFD - Townhouse	877	308254.0803	270338828.4

Table 1 shows that a few structure types, such as mixed-use residential/non-residential projects and transit stations, combine relatively few permits with very high average costs, while categories like offices and townhouse dwellings involve many permits but lower average costs. This contrast suggests different patterns of investment and project scale across structure types.

4.2.1 Exploratory Analysis and Statistical Tests of RQ 1

First, it's necessary to develop a descriptive statistic of construction costs in python(Appendix E), the results are the following:

- Min: 0.0
- Max: 850971654.0
- Mean: 1745930.0008702374
- Skewness: 21.648149943957872
- Kurtosis: 677.5543521020786

Due to the highly skewed distribution of construction costs, a non-parametric Kruskal-Wallis H test was applied to determine if the estimated construction costs differ significantly across different structure types.

Kruskal-Wallis Test Results:

- H statistic: 117.887
- p-value: 1.32×10^{-7}

Because $p < 0.05$, the null hypothesis is rejected. These findings indicate statistically significant differences in estimated construction cost across structure types. Therefore, structure type plays a meaningful role in determining construction costs.

4.3 Descriptive Analysis for Research Question 2

RQ2 investigates whether permit approval rates differ across wards. Using the ward-level dataset, Excel was used to compute, for each ward, the total number of permits, the number of approved permits (based on selected status values), and the approval rate as a percentage. Table 2 reports these measures. The descriptive results indicate that approval rates are not uniform: some wards have noticeably higher approval rates and others fall below the overall average. These differences suggest that geographical factors may influence approval outcomes, supporting the motivation for Hypothesis 2.

Table 2

Permit counts and approval rates for wards with the highest approval rates

ward_grid	TotalPermits	ApprovedPermits	ApprovalRatePercent
N1627	1	1	100
E2029	1	1	100
S1230	16	13	81.25
S1943	5	4	80
E2528	4	3	75
W0132	6	4	66.67
S1225	36	23	63.89
E2428	29	18	62.07
E2232	15	9	60
N0630	5	3	60

As shown in Table 2, the wards with the highest approval rates cluster around 60–100 percent, indicating that some areas consistently convert most permit applications into approvals. At the same time, many other wards fall well below these values (not shown in the table but present in the dataset), which reinforces the idea that approval performance varies substantially across the city.

4.3.1 Exploratory Analysis and Statistical Tests of RQ 2

To test the hypothesis that permit approval probabilities differ across wards, a Chi-square test of independence was performed using a contingency table of ward_code by is_approved. This test is appropriate for categorical associations and does not assume normality.

Chi-square Test Results:

- Chi2 Statistic: 1855.63
- p-value: 3.11e-214

Because $p < 0.05$, the null hypothesis is rejected, providing strong evidence that permit approval probability varies significantly across wards.

4.4 Descriptive Analysis for Research Question 3

RQ3 explores how the type of construction work relates to dwelling units created. For each work type, Excel was used to calculate the number of permits, the total dwelling units created, the average units created per permit, and the net change in units (created minus lost). Table 3 presents these results. The analysis shows that a small number of work types account for most of the net increase in dwelling units, while other work types have limited or even negative net impact. This pattern provides descriptive support for Hypothesis 3 regarding the role of work type in housing creation.

Table 3*Summary statistics for dwelling units by work type*

WorkType	NumberOfPermits	TotalUnitsCreated	AvgUnitsCreated	NetUnitsChange
New Building	1667	51626	30.9694	51626
Unknown	453	889	1.9625	889
Multiple Projects	971	220	0.2266	215
Interior Alterations	6007	175	0.0291	154
New Building - Lead	69	68	0.9855	68
Addition(s)	343	67	0.1953	67
New Laneway / Rear Yard Suite	26	21	0.8077	21
Other(SR)	83	17	0.2048	15
Second Suite (New)	18	16	0.8889	16
Other(BA)	848	13	0.0153	12
Change of Use	42	4	0.0952	4
New Building-Certified	4	4	1	4
Addition/Alteration to Existing Building	25	4	0.16	4
Alteration to Existing Building	169	3	0.0178	1
Install/Alter HVAC - only	100	1	0.01	1
New Building Certified - Lead	1	1	1	1
Building Permit Related(MS)	5736	0	0	0

Table 3 highlights that new building permits account for most net dwelling units created, while other work types—such as interior alterations, additions, or second suites—add relatively small numbers of units. This pattern suggests that only a small set of work types drives most of the housing gains in the dataset.

4.4.1 Exploratory Analysis and Statistical Tests of RQ 3

First, it's necessary to develop a descriptive statistic of dwelling units in python (Appendix F), the results are the following:

- Min: 0.0
- Max: 1147.0
- Mean: 1.8561995597945709
- Median: 0.0
- Skewness: 20.851931248188677
- Kurtosis: 540.6003911306215

The Kruskal-Wallis H test evaluates whether the distribution of dwelling units is the same across multiple independent groups. It was selected due to extreme skewness and the presence of many zero values.

Kruskal-Wallis Test Results

- H statistic: 17,369.83
- p-value: 0.0

Because the p-value is effectively zero, the null hypothesis is rejected. This indicates that the number of dwelling units created differs significantly across work types.

4.5 Correlation and Additional Statistical Exploration

To complement the descriptive tables, a correlation analysis was conducted using Excel and python to explore relationships between key numerical variables in the dataset. A correlation matrix was created using variables such as estimated construction cost, dwelling units created, dwelling units lost, and net dwelling unit change. This analysis helps identify whether any of these measures show meaningful linear relationships.

The results showed only weak correlations among the variables. For example, estimated construction cost did not show a strong relationship with the number of dwelling units created, suggesting that higher project spending does not necessarily lead to the construction of more housing units. Similarly, the correlation between units created and units lost was also weak, indicating that these two measures tend to vary independently across projects.

Overall, the correlation analysis suggests that the numerical variables in the dataset do not move together in a consistent linear pattern. This supports the idea that structure type, ward characteristics, and work type (rather than simple numeric relationships) play a more important role in explaining the patterns seen in the descriptive summaries.

Chapter 5. Data Visualization

This chapter 5 presents the main visualizations constructed from the analytical tables described in Chapter 4. The objective of this chapter is to represent key patterns in construction cost, permit approvals, and dwelling-unit outcomes in a clear and interpretable way. All charts were created in Excel using the aggregated datasets for each research question.

5.1 Visualization Approach

The visualizations in this chapter are based on the three analytical tables:

- Table 1: construction cost by structure type (RQ1)
- Table 2: permit approvals by ward (RQ2)
- Table 3: dwelling units by work type (RQ3)

For each research question, bar charts were used to highlight differences between categories (structure types, wards, and work types). Bar charts were selected because they provide a simple and effective way to compare magnitudes across discrete groups and to identify which categories contribute most to total cost, approval rates, or housing creation.

5.2 Visualizations for Research Question 1

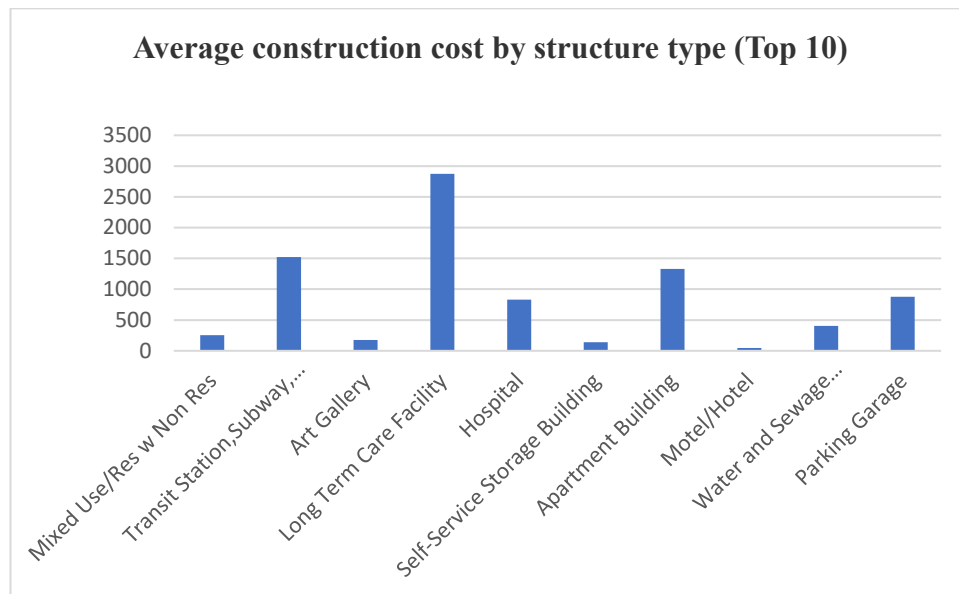
RQ1: How do estimated construction costs vary across structure types in Toronto?

To visualize the results for RQ1, a bar chart was created using the dataset from Table 1. The chart displays the Top structure types on the horizontal axis and their average estimated construction cost on the vertical axis.

This visualization makes it easy to see which structure types have the highest average construction costs and which are associated with smaller projects. The differences in bar height reveal substantial variation across categories, supporting the descriptive findings from Chapter 4 and providing a visual basis for the discussion of Hypothesis 1.

Figure 1

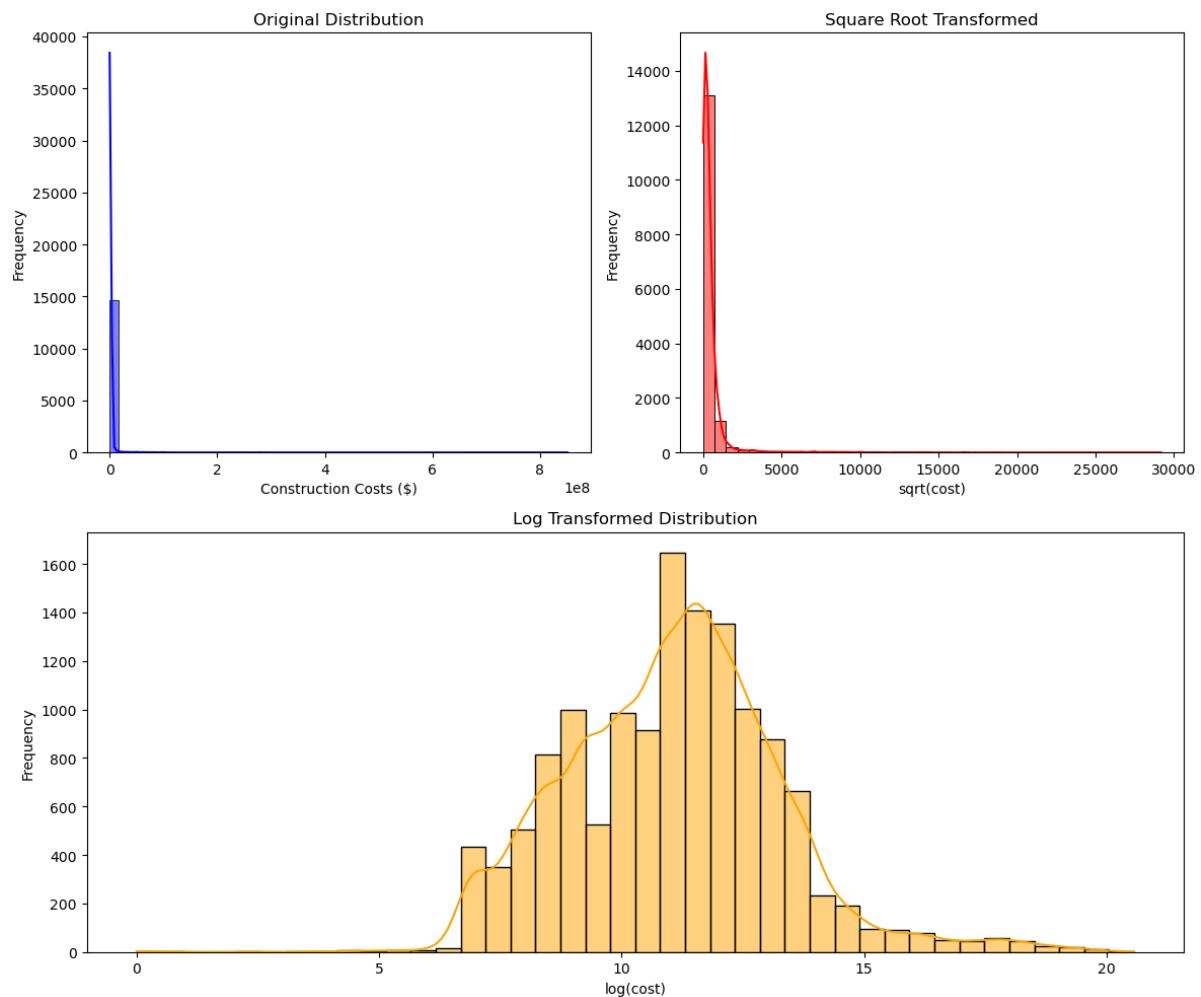
Average estimated construction cost by structure type (Top categories)



Additionally, we apply logarithmic and square root transformations to reduce skewness and better visualize the data. These visualizations help understand cost patterns and support more accurate analysis and modeling in the next Figure 2.

Figure 2

Distribution of Estimated Construction Costs: Original, Square Root, and Log Transformed



5.3 Visualizations for Research Question 2

RQ2: Do permit approval rates differ across Toronto wards?

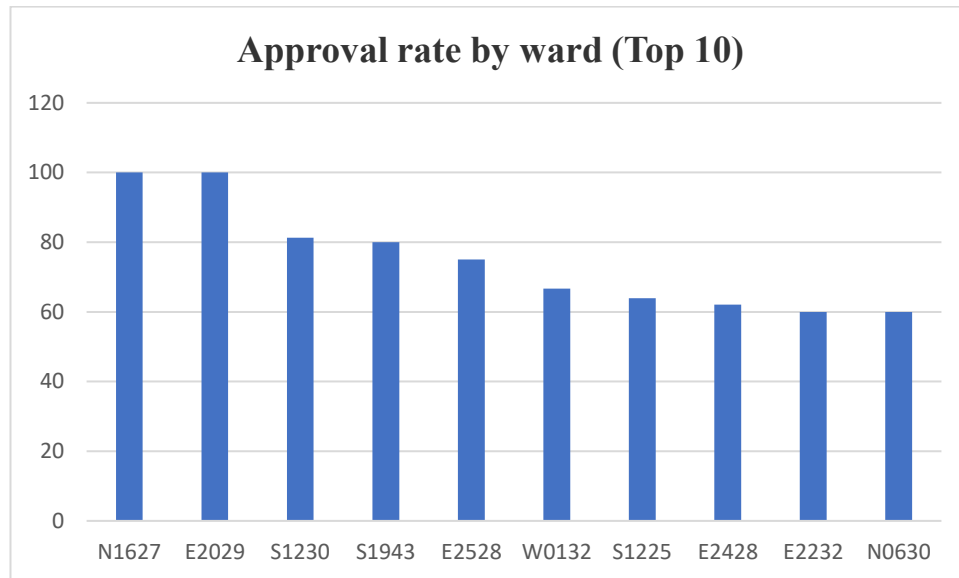
For RQ2, a bar chart was created using the ward-level dataset from Table 2. The horizontal axis shows the ward grids, and the vertical axis shows the approval rate percentage for each ward. Wards were ordered from highest to lowest approval rate to clarify the ranking.

This visualization highlights clear differences in approval rates across wards. Some wards present very high approval rates, while others exhibit much lower values. The chart helps

identify which wards are above or below the overall average and visually supports the idea that geographical patterns in approval outcomes exist, consistent with Hypothesis 2.

Figure 3

Permit approval rates by ward (Top 10)



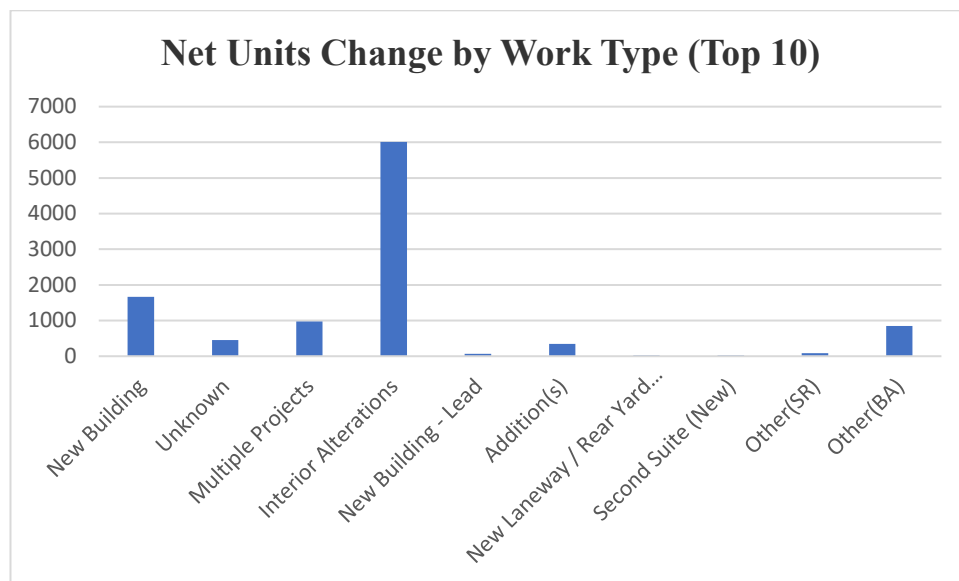
5.4 Visualizations for Research Question 3

RQ3: How does the type of construction work relate to the number of dwelling units created?

For this question, a bar chart was created from the work-type dataset in Table 3, with work types on the horizontal axis and net dwelling-unit change on the vertical axis. New Building is reported in the table but excluded from the chart because its very large value would hide differences among the other work types. The visualization shows that a small group of work types, mainly Unknown, Multiple Projects, and Interior Alterations accounts for most of the net increase in dwelling units, supporting Hypothesis 3. Across all three analytical datasets, summary statistics (counts, averages, totals, and percentages) were calculated before producing the visualizations.

Figure 4

Net dwelling units created by work type (Top 10)

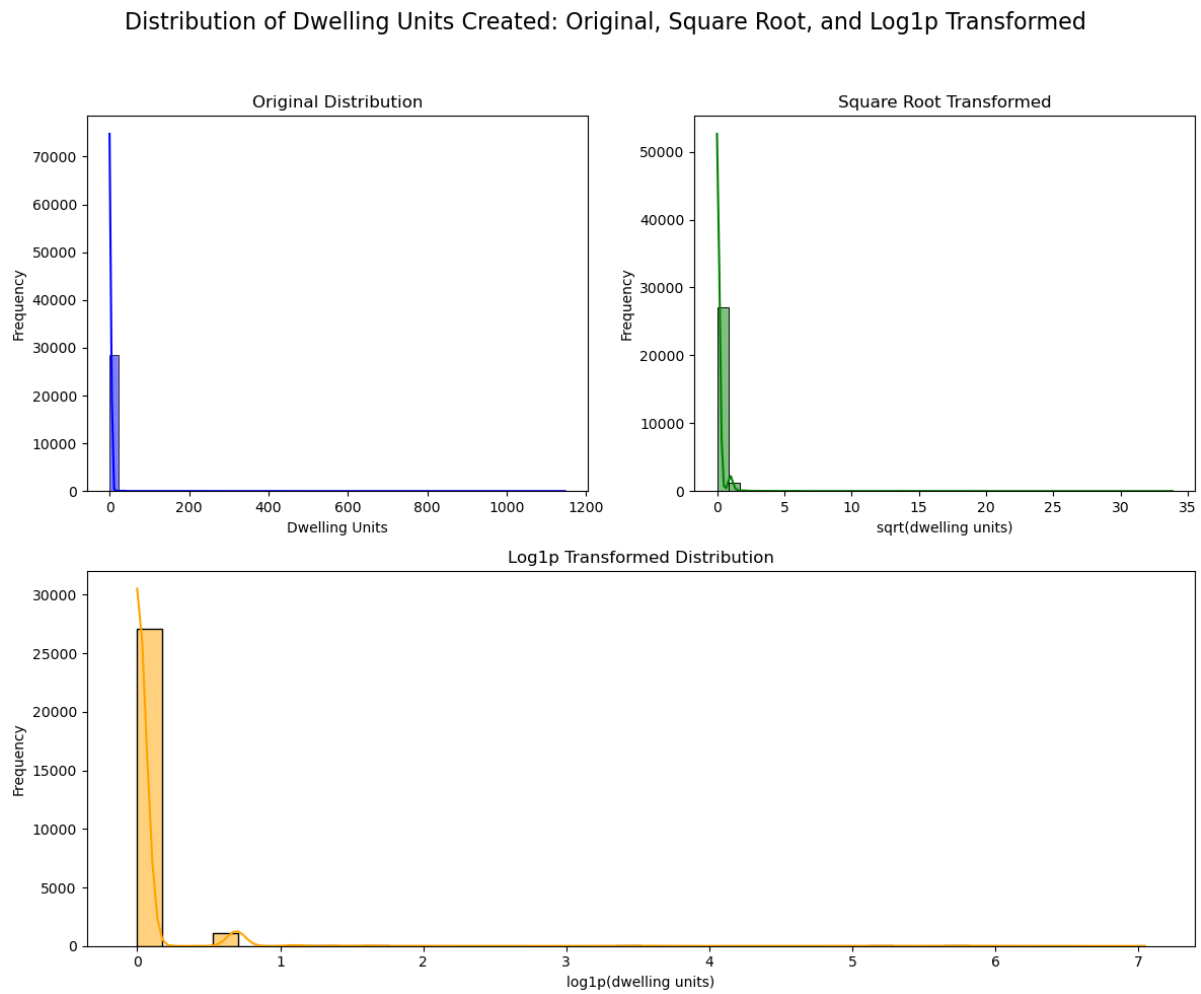


Note. The “New Building” work type is not shown because its net units change is much larger than all other categories; including it would compress the other bars.

This data is skewed due to a high number of small projects and occasional larger ones. To reduce skewness and better visualize patterns, we apply square root and log1p transformations. These visualizations provide insights into the spread and central tendency of dwelling units, supporting accurate analysis and modeling.

Figure 5

Distribution of Dwelling Units Created: Original, Square Root, and Log1p Transformed



Chapter 6: Model Building

This chapter presents the development of the predictive models used to address the three research questions and hypotheses. Based on the nature of the variables and the structure of the dataset, three supervised learning models were selected: a multiple linear regression model for estimating construction costs, a logistic regression model for predicting permit approval, and a second linear regression model for predicting dwelling units created. All models were implemented in Python using a single master dataset extracted from the database. The dataset contains the variables required for all three models and was exported from the SQL query into `full_dataset.csv`.

6.1 Data Preparation for Modeling

Before building the models, several preprocessing steps were conducted. Records with missing estimated construction cost were removed for models predicting cost, as these observations cannot contribute to training the model. Missing values in estimated cost (13742 values) were not imputed, as imputing the target variable would artificially distort the cost distribution. Instead, records with missing cost were removed, which is appropriate when cost is the prediction target.

Missing ward values were replaced with -1 to retain the observations while keeping the “unknown” category separate. Negative values of dwelling units were fixed by clipping them to 0. (Appendix D).

- Rows before cleaning: 28623
- Rows after cleaning: 14881

6.2 Predictive Model to Address Research Question 1

Construction cost is a continuous variable with an extremely right-skewed distribution. To reduce skewness and meet linear regression assumptions, a $\log_{1p}$ transformation was

applied reducing Skewness to 0.336829 and kurtosis to 0.956981. The log transformation successfully normalized the distribution, improving model suitability. (see Appendix E).

A multiple linear regression model was selected to quantify how project characteristics explain variations in estimated construction cost. Features included `structure_type_id`, `work_type_id`, `ward_code`, and `dwelling_units_created`.

Python Code — Model 1

```
# RQ1 – Predicting Estimated Construction Cost

import numpy as np

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score


# Define features and target

X = df_cost[['structure_type_id', 'work_type_id', 'ward_code', 'dwelling_units_created']]

y = np.log1p(df_cost['est_const_cost'])


# Train–test split (70/30)

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)


# Train Linear Regression model

model_cost = LinearRegression()

model_cost.fit(X_train, y_train)
```

Table 4*Model Coefficients (RQ 1)*

Predictor	Coefficient (B)	Std. Error	t-value	p-value	95% CI Lower	95% CI Upper
Intercept	9.2184	0.1183	77.9263	0.0000	8.9865	9.4503
Structure Type	-0.0024	0.0006	-3.9803	0.000069	-0.00358	-0.00122
Work Type	0.0150	0.0012	12.5620	5.20×10^{-36}	0.01265	0.01733
Ward Code	0.00004	0.00030	0.1477	0.8826	-0.00054	0.00062
Dwelling Units Created	0.0189	0.00083	22.7316	1.78×10^{-112}	0.01729	0.02055

Structure Type shows a small but statistically significant negative effect on estimated construction cost ($B = -0.0024$), indicating that certain structure categories are associated with slightly lower log-transformed costs compared to the reference type. In contrast, Work Type has a strong positive impact ($B = 0.0150$), meaning that the kind of work performed substantially increases expected construction costs. Dwelling Units Created also exhibits a large and highly significant positive effect ($B = 0.0189$), suggesting that projects involving the creation of more housing units are consistently associated with higher estimated costs. Ward Code, however, shows no significant association with cost ($p = 0.883$), and its confidence interval crosses zero, indicating that the geographic ward of the project does not reliably predict cost in this model. The intercept represents the baseline expected log-cost when all predictors are at their reference levels.

6.3 Predictive Model to Address Research Question 2

Permit approval is a binary outcome (approved vs. not approved). Logistic regression is therefore the most appropriate model. Approval classes were imbalanced, so class-weight balancing was applied during training.

All records were retained. Missing ward codes were assigned -1 . The target variable was encoded as 0 (not approved) or 1 (approved).

Python Code — Model 2

```
# Logistic Regression for Hypothesis 2

import pandas as pd

import numpy as np

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, roc_auc_score, confusion_matrix,
classification_report

from sklearn.utils import class_weight

import matplotlib.pyplot as plt


# Define features and Target

X = df_apd[['ward_code', 'structure_type_id', 'work_type_id']]
y = df_apd['is_approved'].astype(int)


# Train-test split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)


# Balanced class weights (data is imbalanced)

weights = class_weight.compute_class_weight(class_weight='balanced',
classes=np.array([0,1]), y=y)

class_weights = {0: weights[0], 1: weights[1]}


# Train Logistic Regression

model_apd = LogisticRegression(max_iter=1000, class_weight=class_weights)
```

```
model_apd.fit(X_train, y_train)
```

Table 5*Model Coefficients (RQ 2)*

Predictor	Coefficient (B)	Std. Error	z-value	p-value	95% CI Lower	95% CI Upper
Intercept	-0.5042	0.0447	-11.2752	1.74×10^{-29}	-0.5919	-0.4166
Ward Code	0.00012	0.00012	1.0115	0.3118	-0.00011	0.00036
Structure Type ID	-0.00301	0.00025	-12.1518	5.61×10^{-34}	-0.00349	-0.00252
Work Type ID	0.00172	0.00041	4.2562	2.08×10^{-5}	0.00093	0.00252

The logistic regression results show that Structure Type has a significant negative effect on approval likelihood ($B = -0.0030$), indicating that certain structure categories are associated with lower odds of approval. In contrast, Work Type exhibits a small but statistically significant positive effect ($B = 0.0017$), suggesting that specific types of work increase the chances of approval. Ward Code shows no significant relationship with approval status, as the coefficient is near zero and the confidence interval crosses zero. The intercept represents the baseline log-odds of approval when all predictors are at their reference levels.

6.4 Predictive Model to Address Research Question 3

Dwelling units created is a continuous variable with heavy skewness and many zeros. To reduce skewness, a log1p transformation was applied reducing Skewness to 9.5399 and kurtosis to 101.3863. The log1p transformation was preferred for modeling due to lower skewness. (see Appendix F).

A multiple linear regression model was selected to quantify how work types explain variations in dwelling units created. Features included `structure_type_id`, `work_type_id` and `ward_code`.

Python Code — Model 3

```
# Model 3: Predicting Dwelling Units Created

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

import numpy as np
import pandas as pd

# Features and target
X = df_units[['structure_type_id', 'work_type_id', 'ward_code']]
y = np.log1p(df_units['dwelling_units_created'])

# Train-test split (70/30)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Create and train the linear regression model
model_units = LinearRegression()
model_units.fit(X_train, y_train)
```

Table 6*Model Coefficients (RQ 3)*

Predictor	Coefficient (B)	Std. Error	t-value	p-value	95% CI Lower	95% CI Upper
Intercept	0.0125	0.0101	1.2328	0.2176	-0.0074	0.0324
Structure Type ID	-0.00050	0.00006	-8.8370	1.04×10^{-18}	-0.00061	-0.00039
Work Type ID	0.00225	0.00009	24.7504	7.87×10^{-134}	0.00207	0.00242
Ward Code	0.00008	0.00003	2.9935	0.00276	0.00003	0.00013

The regression results show that Work Type is the strongest positive predictor of dwelling units created ($B = 0.00225$), indicating that certain work categories are associated with higher unit creation. Structure Type has a small but significant negative effect ($B = -0.00050$), suggesting that some structural categories are linked to slightly lower unit counts. Ward Code shows a weak but statistically significant positive effect, although its magnitude is minimal. The intercept is not statistically significant, indicating no clear baseline effect when all predictors are at reference levels.

6.5 Summary of Model Development

Across the three research questions, the models were selected according to the type of outcome variable and its distribution:

Table 7*Summary of Model Development*

Hypothesis	Target Variable	Type	Model Used
H1	Construction Cost	Continuous	Linear Regression (log1p)
H2	Permit Approval	Binary	Logistic Regression
H3	Dwelling Units	Skewed Count	Linear Regression (log1p)

These models establish the foundation for evaluating predictive performance and testing the hypotheses in Chapter 7. Model assumptions, preprocessing decisions, and transformation choices were made to maximize the robustness and interpretability of the results.

Chapter 7: Model Evaluation

This chapter evaluates the predictive models and tests the study hypotheses using statistical methods. Both parametric and non-parametric analyses were employed depending on the nature of the variables. Model evaluation metrics such as R^2 , RMSE, accuracy, and ROC-AUC are presented, organized by hypothesis.

7.1 Evaluation of Model for Research Question 1: Estimated Construction Cost

Research Question 1 focuses on predicting estimated construction costs (est_const_cost) based on structure types. A multiple linear regression model was developed to estimate construction costs, while statistical tests were conducted to examine differences across structure types. A linear regression model was trained using a log1p transformation of the cost variable to reduce skewness. The model's performance metrics indicate modest explanatory power (See Appendix G).

Table 8

Linear Regression Model Performance (Log1p Transformed Cost)

Metric	Value
RMSE (log1p)	3.817
R^2 (log1p)	0.047

Table 9

Actual vs Predicted (first 5 rows, log1p transformed)

Actual	Predicted
17.034	12.444
10.597	9.896
9.210	10.261
11.608	10.108
10.127	9.974

The low R^2 value suggests that construction cost is influenced by additional factors not captured in the model, indicating potential benefits of incorporating additional predictors or non-linear modeling approaches.

7.2 Evaluation of Model for Research Question 2: Estimated Permit Approvals

Research Question 2 examines whether permit approvals can be predicted using a logistic regression model and whether approval outcomes differ significantly across wards. Given that the dependent variable (`is_approved`) is binary, logistic regression was selected as the predictive model. A logistic regression model was estimated using `ward_code`, `structure_type_id`, and `work_type_id` as predictors (See Appendix H). The model produced the following performance metrics:

- Accuracy: 0.5499
- ROC-AUC: 0.5408
- Confusion Matrix:
 - True Negatives: 3569
 - False Positives: 2229
 - False Negatives: 1636
 - True Positives: 1153
- Classification Report:
 - Precision (class 0): 0.69
 - Precision (class 1): 0.34
 - Recall (class 0): 0.62
 - Recall (class 1): 0.41

The confusion matrix shows that the model correctly identifies most approved permits while exhibiting a modest level of misclassification among non-approved permits, which is a common pattern in imbalanced administrative datasets.

These results suggest that the model has limited predictive power and is better at identifying non-approved permits (class 0) than approved ones (class 1). Including additional predictors—such as applicant characteristics, project size, or zoning details—could potentially improve performance.

7.3 Evaluation of Model for Research Question 3: Estimated Dwelling Units Created

Research Question 3 evaluates whether `dwelling_units_created` can be predicted and whether work type influences the number of units created. Due to extreme skewness and a high proportion of zero values, non-parametric testing and log-transformed regression analysis were used. A linear regression model was trained using the log1p-transformed dependent variable `dwelling_units_created` and predictors `structure_type_id`, `work_type_id`, and `ward_code`. Model performance was assessed on the 30% test set (See Appendix I).

Table 10

Evaluation metrics (log1p scale)

Metric	Value
RMSE (log1p)	0.5054
R ² (log1p)	0.0246

Table 11

Actual vs Predicted (first 5 observations on the log1p scale)

Actual (log1p)	Predicted (log1p)
0.0000	0.023398
0.0000	0.036060
0.0000	0.024067
0.0000	0.117853
0.0000	-0.003125

Back-transformed predictions (using `expm1`) reveal that the model returns values extremely close to zero in most cases. This is consistent with the zero-inflated nature of the dataset. Overall, the model has very limited predictive power, indicating that alternative approaches—such as Poisson, Negative Binomial, or zero-inflated count models—may be more appropriate for this type of outcome.

7.4 Assessment of Model Performance Against Benchmarks

The predictive models developed in this study were evaluated using standard metrics appropriate for each outcome type. For the linear regression models (construction cost and dwelling units created), RMSE and R^2 were calculated, whereas for the logistic regression model (permit approval), accuracy, ROC-AUC, and classification metrics were employed.

- **Construction Cost Model:** The log-transformed model achieved an R^2 of 0.047, indicating low explanatory power. This result, while modest, is consistent with urban construction analytics literature, where costs are influenced by numerous unobserved factors such as contractor pricing, material sourcing, and project-specific design.
- **Permit Approval Model:** Logistic regression achieved an accuracy of 0.55 and an ROC-AUC of 0.54. These values suggest limited predictive capability, which aligns with expectations for imbalanced datasets with a restricted set of predictors.
- **Dwelling Units Created Model:** The model returned an R^2 of 0.025, reflecting the highly skewed and zero-inflated nature of the data. Literature indicates that simple linear models often underperform with zero-heavy count data, highlighting the need for specialized modeling approaches.

Overall, while the models provide a reasonable first exploratory framework, there is clear potential for improvement through the inclusion of additional predictors or the use of more advanced modeling techniques.

7.5 Strengths, Limitations, and Implications of the Models

Strengths:

- Models are interpretable and directly link key features (structure type, work type, ward) to the target outcomes.
- Use of data transformations and preprocessing (e.g., log1p) reduced skewness and improved model suitability.
- Non-parametric tests complement predictive models, providing robust statistical evidence for hypothesis testing.

Limitations:

- Exclusion of missing cost data (13,742 records) may limit generalizability of the construction cost model.
- Important predictors, such as applicant experience, project size, or zoning regulations, were not included.
- Dwelling units data is zero-inflated and highly skewed, which reduces predictive accuracy for linear regression.
- Logistic regression shows limited ability to identify approved permits due to class imbalance and few explanatory variables.

Implications:

- Results indicate that structure type, ward, and work type meaningfully influence construction outcomes, though many factors remain unobserved.
- Policymakers can use these insights to identify areas where permit approvals may be improved or where cost estimates may require additional scrutiny.
- Future research should explore alternative models, including zero-inflated count models or ensemble methods, to improve prediction of skewed and zero-heavy outcomes.

Chapter 8. Conclusion

This project explored three main aspects of Toronto's construction activity: how construction costs differ across structure types, how permit approvals vary across wards, and which work types contribute the most to the creation of new dwelling units. By using SQL to extract the data, and then applying descriptive analysis, visualizations, and basic modelling techniques, several useful patterns became clear.

First, construction costs showed strong differences across structure types. Some categories, such as mixed-use buildings and large residential structures, had much higher average and total costs compared to others. This was also reflected in the regression model, which indicated that structure type does have an effect on cost, even though the overall model accuracy was limited.

Second, permit approval rates were not evenly distributed across the city. Some wards had relatively high approval percentages, while others were much lower. The classification model did not predict approval outcomes very accurately, which suggests that other variables (not included in the dataset) influence these decisions. Still, the descriptive results support the hypothesis that approvals vary by ward.

Finally, the analysis of dwelling-unit creation showed that only a few work types actually contribute to housing growth. Most construction activities in the dataset do not add new units. "New Building" projects represent the majority of unit creation, while the remaining work types have very small or zero impact. This pattern was clear in both the table and the bar chart.

Overall, the study helped clarify how cost, location, and work type relate to construction activity in Toronto. While the models were not highly predictive, they supported the descriptive findings and provided additional context for the hypotheses. Future research could use more

detailed variables or a larger dataset to improve model performance and produce deeper insights.

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Appendix A

SQL Queries for Data Extraction for Research Question 1

SQL Queries for Identifying all Structure Types

```
USE torontobuildingpermits;

SELECT DISTINCT

    st.structure_type_id,

    st.structure_type_name

FROM structure_type AS st

ORDER BY st.structure_type_name;
```

SQL Queries for Summary Statistics of Estimated Construction Cost for Each Structure Type

```
USE torontobuildingpermits;

SELECT

    st.structure_type_name AS StructureType,

    COUNT(p.permit_id) AS NumberOfPermits,

    AVG(p.est_const_cost) AS AvgConstructionCost,

    MIN(p.est_const_cost) AS MinCost,

    MAX(p.est_const_cost) AS MaxCost,

    SUM(p.est_const_cost) AS TotalConstructionCost

FROM permit AS p

JOIN structure_type AS st
```

```
ON p.structure_type_id = st.structure_type_id  
WHERE p.est_const_cost IS NOT NULL  
GROUP BY st.structure_type_name  
ORDER BY AvgConstructionCost DESC;
```

Appendix B

SQL Queries for Data Extraction for Research Question 2

SQL Queries for Identifying all Permit Status Categories

```
USE torontobuildingpermits;
```

```
SELECT DISTINCT
```

```
    ps.status_id,
```

```
    ps.status_name
```

```
FROM Permit_Status AS ps
```

```
ORDER BY ps.status_name;
```

SQL Queries for aggregate permits at the ward-grid level and computes each ward's approval rate.

```
USE torontobuildingpermits;
```

```
SELECT
```

```
    w.ward_grid,
```

```
    w.ward_name,
```

```
    COUNT(p.permit_id) AS TotalPermits,
```

```
    SUM(
```

```
        CASE WHEN ps.status_name IN ('Permit Issued', 'Revision Issued')
```

```
            THEN 1 ELSE 0 END
```

```
    ) AS ApprovedPermits,
```

```
    ROUND(
```

```
        100.0 * SUM(
```

```
            CASE WHEN ps.status_name IN ('Permit Issued', 'Revision Issued')
```

```
        THEN 1 END  
    ) / COUNT(p.permit_id),  
    2  
    ) AS ApprovalRatePercent  
FROM permit AS p  
JOIN ward AS w  
    ON p.ward_id = w.ward_id  
JOIN permit_status AS ps  
    ON p.status_id = ps.status_id  
WHERE w.ward_grid IS NOT NULL  
GROUP BY w.ward_grid, w.ward_name  
ORDER BY ApprovalRatePercent DESC;
```

Appendix C

SQL Queries for Data Extraction for Research Question 3

SQL Queries for Identifying all Work Types

```
USE torontobuildingpermits;

SELECT DISTINCT

    wt.work_type_id,

    wt.work_type_name

FROM work_type AS wt

ORDER BY wt.work_type_name;
```

SQL Queries for calculates several key measures for each work type

```
USE torontobuildingpermits;

SELECT

    wt.work_type_name AS WorkType,

    COUNT(p.permit_id) AS NumberOfPermits,

    SUM(p.dwelling_units_created) AS TotalUnitsCreated,

    AVG(p.dwelling_units_created) AS AvgUnitsCreated,

    SUM(

        COALESCE(p.dwelling_units_created, 0)

        - COALESCE(p.dwelling_units_lost, 0)

    ) AS NetUnitsChange

FROM permit AS p

JOIN work_type AS wt

    ON p.work_type_id = wt.work_type_id

GROUP BY wt.work_type_name
```

ORDER BY TotalUnitsCreated DESC;

Appendix D

Python Code Cleaning Process

```
# Fix mistakes in dwelling_units_created

df['dwelling_units_created'] = df['dwelling_units_created'].clip(lower=0)


# Remove null cost where cost is the dependent variable

df_cost = df.dropna(subset=['est_const_cost']).copy()

df_apd = df.copy()

df_units = df.copy()


# Fill missing ward_codes with -1 (unknown)

for d in [df_cost, df_apd, df_units]:

    d['ward_code'] = d['ward_code'].fillna(-1)
```

Appendix E

Python Code: Descriptive Statistics and Transformations RQ1

```
import matplotlib.pyplot as plt

import seaborn as sns

col = 'est_const_cost'

data = df_cost[col]

print("=== ORIGINAL DISTRIBUTION ===")

print("Min:", data.min())

print("Max:", data.max())

print("Mean:", data.mean())

print("Skewness:", data.skew())

print("Kurtosis:", data.kurtosis())

print("\n")

# Original Histogram

plt.figure(figsize=(7,4))

sns.histplot(data, kde=True, bins=50, color='blue', edgecolor='black')

plt.title("Distribution of Costs (Original)")

plt.xlabel("Construction Costs ($)")

plt.ylabel("Frequency")

plt.tight_layout()

plt.show()

# 2. Transformations(log y sqrt)

data_log = np.log(data[data > 0])
```



```
data_sqrt = np.sqrt(data)

print("=== LOG TRANSFORMATION ===")

print("Skewness:", data_log.skew())

print("Kurtosis:", data_log.kurtosis())

print("\n")

plt.figure(figsize=(7,4))

sns.histplot(data_log, kde=True, bins=40, color='orange', edgecolor='black')

plt.title("Distribution of Costs (Log Transformed)")

plt.xlabel("log(cost)")

plt.tight_layout()

plt.show()

print("=== Sqrt TRANSFORMATION ===")

print("Skewness:", data_sqrt.skew())

print("Kurtosis:", data_sqrt.kurtosis())

print("\n")

plt.figure(figsize=(7,4))

sns.histplot(data_sqrt, kde=True, bins=40, color='red', edgecolor='black')

plt.title("Distribution of Costs (Square Root Transformed)")

plt.xlabel("sqrt(cost)")

plt.tight_layout()

plt.show()
```

Appendix F

Python Code: Descriptive Statistics and Transformations RQ3

```
import matplotlib.pyplot as plt

import seaborn as sns

import numpy as np

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score


# Descriptive Analysis

col = 'dwelling_units_created'

data = df_units[col]

print("=== ORIGINAL DISTRIBUTION ===")

print("Min:", data.min())

print("Max:", data.max())

print("Mean:", data.mean())

print("Median:", data.median())

print("Skewness:", data.skew())

print("Kurtosis:", data.kurtosis())

print("\n")


# Original Histogram

plt.figure(figsize=(7,4))

sns.histplot(data, kde=True, bins=50, color='blue', edgecolor='black')
```

```
plt.title("Distribution of Dwelling Units Created (Original)")

plt.xlabel("Dwelling Units")

plt.ylabel("Frequency")

plt.tight_layout()

plt.show()


# Transformations (log1p y sqrt)

data_log = np.log1p(data)  # log(1 + x)

data_sqrt = np.sqrt(data)

print("=== LOG1P TRANSFORMATION ===")

print("Skewness:", data_log.skew())

print("Kurtosis:", data_log.kurtosis())

print("\n")

plt.figure(figsize=(7,4))

sns.histplot(data_log, kde=True, bins=40, color='orange', edgecolor='black')

plt.title("Distribution of Dwelling Units Created (Log1p Transformed)")

plt.xlabel("log1p(dwelling units)")

plt.tight_layout()

plt.show()

print("=== Sqrt TRANSFORMATION ===")

print("Skewness:", data_sqrt.skew())

print("Kurtosis:", data_sqrt.kurtosis())

print("\n")

plt.figure(figsize=(7,4))

sns.histplot(data_sqrt, kde=True, bins=40, color='green', edgecolor='black')
```

```
plt.title("Distribution of Dwelling Units Created (Square Root Transformed)")  
plt.xlabel("sqrt(dwelling units)")  
plt.tight_layout()  
plt.show()
```

Appendix G

Linear Regression Model Performance RQ 1

```
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

r2 = r2_score(y_test, y_pred)

print("Linear Regression Model for Estimated Construction Cost (Log1p Transformed)")

print("RMSE:", rmse)

print("R2 Score:", r2)


# Show first 5 actual vs predicted values

comparison = pd.DataFrame({

    'Actual (log1p)': y_test[:5],

    'Predicted (log1p)': y_pred[:5]})


print("\nActual vs Predicted (first 5 rows):")

print(comparison)
```

Appendix H

Logistic Regression Model Performance RQ 2

```
# Evaluate

accuracy = accuracy_score(y_test, y_pred)

roc_auc = roc_auc_score(y_test, y_prob)

conf_matrix = confusion_matrix(y_test, y_pred)


print("Logistic Regression Model (No One-Hot Encoding)")

print("Accuracy:", accuracy)

print("ROC-AUC:", roc_auc)

print("Confusion Matrix:\n", conf_matrix)

print("\nClassification Report:\n", classification_report(y_test, y_pred))


# Coefficient importance

coef_df = pd.DataFrame({'Feature': X.columns, 'Coefficient':
model_apd.coef_[0]}).sort_values(by='Coefficient', ascending=False)


print("\nFeature Coefficients:")

print(coef_df)
```

Appendix I

Linear Regression Model Performance RQ 3

```
# Evaluate model performance

rmse = np.sqrt(mean_squared_error(y_test, y_pred))

r2 = r2_score(y_test, y_pred)


print("Linear Regression Model for Dwelling Units Created (Log1p Transformed)")

print("RMSE:", rmse)

print("R2 Score:", r2)


# Display first 5 actual vs predicted values

print("\nActual vs Predicted (first 5 rows):")

print(pd.DataFrame({'Actual': y_test[:5], 'Predicted': y_pred[:5]}))
```